Math Inquirers Level 2: Scheme of Work

Holiday program for children whose school year covers Primary 2 topics • 10 lessons of 90 minutes • 2 lessons a week for 5 weeks • live online, up to 8 children

Approach. Enrichment, not extra drill: non-routine puzzles tagged to the school syllabus. Every lesson follows the same routine: warm-up (10 min), explore with objects or drawings (15), mini-lesson (10), brain break (5), worksheet in pairs then alone (30), talk time (10), exit ticket and journal stamp (10).
Habits. Notice it. Test it. Explain why.

Lesson	Title and syllabus topic	Objective	Skills and heuristics practised	Exit ticket
1 Week 1A	**Digit Detectives** Numbers to 1000	Read and build 3-digit numbers by place value (also with more than 9 tens), make every number from a set of digit cards by listing in order, find the biggest and smallest, and use patterns on a 1000 chart to count on and back in 1s, 10s and 100s.	Place value, making a list in order, looking for a pattern, reasoning about zero	Use the digit cards 6, 0 and 2 once each. Write the biggest and the smallest 3-digit number.
2 Week 1B	**Missing Digit Mysteries** Addition and Subtraction within 1000	Find missing digits in 3-digit addition and subtraction by starting at the ones and checking for a carry, use "=" as "is the same as" to balance number sentences, and complete magic squares and triangles with bigger numbers.	Working backwards, guess and check, reasoning about regrouping, balancing, drawing a model	3A4 + 257 = 601. Which digit is A?
3 Week 2A	**Legs, Wheels and Arrays** Multiplication and Division	Read an array as rows of equal groups and write two multiplication and two division sentences for it; solve "how many of each" legs and wheels puzzles with a table; share with leftovers and decide what the leftover means; find all the arrays for a number.	Draw it (arrays), make a table, guess and check, working in order to find all the ways	A pen has hens and goats. There are 5 animals and 14 legs. How many hens and how many goats?
4 Week 2B	**Pattern Machines** Patterns with Multiplication	Find far places in a repeating pattern by counting in groups (the ends of the groups are in a times table); count how many of one shape there are; describe a growing pattern with a table and a rule and use it to go forwards and backwards; find the numbers that two counting patterns share.	Look for a pattern, make a table, make a list, explain a rule	The pattern goes triangle, circle, square, triangle, circle, square ... What is the 17th shape? How do you know?
5 Week 3A	**Mystery Number Machines** Working Backwards	Find a mystery number in two-step puzzles with +, −, × and ÷ by working backwards (undo the last step first with the opposite operation), check by going forwards, and use before-and-after bar models for stories where an amount changes.	Inverse operations, working backwards, guess and check, before-and-after bar models, checking forwards	I think of a number. I multiply it by 4, then take away 6. I get 14. What is my number?
6 Week 3B	**Market Day** Money	Read and write money in dollars and cents, pay exactly with the fewest notes and coins, make a list in order to find every way to pay, give change by counting on, decide whether there is enough money, and work backwards to find how much someone had at first.	Systematic listing (make a list in order), working backwards, counting on, comparing, justifying "there are no more ways"	A pen costs $3.65. Pay exactly with the fewest notes and coins.
7 Week 4A	**Fair Shares** Fractions	Recognise equal parts and name fractions with bottom numbers up to 12; compare unit fractions and like fractions; add and take away like fractions within one whole; solve fair-sharing stories with a bar.	Checking for equal parts, comparing with the same whole, drawing bars, explaining	Which is bigger, 1/7 or 1/9? Then work out 2/9 + 5/9.
8 Week 4B	**Shape Builders** 2D and 3D Shapes	Name semicircles and quarter circles and use them to make and break up shapes; count every square and triangle in a figure by working in order; count the cubes in a stack, including hidden ones, layer by layer; decide which nets fold into a cube.	Spatial reasoning, systematic counting, folding in your head, explaining	How many cubes are in a stack with a bottom layer 3 cubes long and 2 cubes deep, and 1 cube on top?

Lesson	Title and syllabus topic	Objective	Skills and heuristics practised	Exit ticket
9 Week 5A	**Measure Masters** Length, Mass, Volume and Time	Solve balance-scale puzzles in g and kg by swapping things that balance; solve pouring puzzles with litre jugs by keeping a table of every pour; join and cut lengths in m and cm (overlaps, pieces and cuts); tell the time to 5 minutes with a.m. and p.m. and find start times, end times and how long, also across the hour.	Reasoning with equal amounts (swapping), working step by step, keeping a table, counting on in time, checking	A lesson starts at 4:50 p.m. and lasts 25 minutes. What time does it end?
10 Week 5B	**Challenge Day** Mixed Puzzles	Choose a way to start a puzzle (draw it, list it in order, make a table) and use it on mixed puzzles: picture graphs with a key of 5 or 10, a logic grid with exactly one answer, numbers to 1000, money, fractions, and × and ÷ with the 2, 3, 4, 5 and 10 tables. Check the answer and explain it.	Choosing a strategy, drawing models, listing in order, making tables, logical elimination, checking, explaining	5 bicycles and tricycles have 13 wheels altogether. How many tricycles are there? How did you start?

Assessment and progress

Each lesson has a 2-page worksheet (Start here), Think harder and Go further pages, an exit ticket and an Inquiry Journal stamp. Tutors record S (secure), D (developing) or H (needs help) for each child on the progress tracker in the tutor guide. Parents receive a short note each week and the Weekly Dose of Thinking Math practice pack.

Materials

Printed student workbook with cut-outs, pencil, scissors and paper; a laptop or tablet with a camera. Tutors use the slides for each lesson and the tutor guide scripts.